

Topic 15 – Forces and matter

Students should:	Maths skills
15.1 Explain, using springs and other elastic objects, that stretching, bending or compressing an object requires more than one force	
15.2 Describe the difference between elastic and inelastic distortion	
15.3 Recall and use the equation for linear elastic distortion including calculating the spring constant: force exerted on a spring (newton, N) = spring constant (newton per metre, N/m) × extension (metre, m) $F = k \times x$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
15.4 Use the equation to calculate the work done in stretching a spring: energy transferred in stretching (joules, J) = 0.5 × spring constant (newton per metre, N/m) × (extension (metre, m)) ² $E = \frac{1}{2} \times k \times x^2$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d 4c, 4e, 4f
15.5 Describe the difference between linear and non-linear relationships between force and extension	4c, 4e
15.6 <i>Core Practical: Investigate the extension and work done when applying forces to a spring</i>	1a, 1c, 1d 2a, 2b, 2c, 2f 3a, 3b, 3c, 3d 4a, 4b, 4c, 4d
15.7P Explain why atmospheric pressure varies with height above the Earth's surface with reference to a simple model of the Earth's atmosphere	
15.8P Describe the pressure in a fluid as being due to the fluid and atmospheric pressure	
15.9P Recall that the pressure in fluids causes a force normal to any surface	
15.10P Explain how pressure is related to force and area, using appropriate examples	1c
15.11P Recall and use the equation: pressure (pascal, Pa) = force normal to surface (newton, N) ÷ area of surface (square metre, m ²) $P = \frac{F}{A}$	1a, 1b, 1c, 1d 2a 3a, 3c, 3d 5b, 5c
15.12P Describe how pressure in fluids increases with depth and density	1c

Students should:	Maths skills
15.13P Explain why the pressure in liquids varies with density and depth	1c
15.14P Use the equation to calculate the magnitude of the pressure in liquids and calculate the differences in pressure at different depths in a liquid: pressure due to a column of liquid (pascal, Pa) = height of column (metre, m) × density of liquid (kilogram per cubic metre, kg/m³) × gravitational field strength (newton per kilogram, N/kg) $P = h \times \rho \times g$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
15.15P Explain why an object in a fluid is subject to an upwards force (upthrust) and relate this to examples including objects that are fully immersed in a fluid (liquid or gas) or partially immersed in a liquid	5b
15.16P Recall that the upthrust is equal to the weight of fluid displaced	
15.17P Explain how the factors (upthrust, weight, density of fluid) influence whether an object will float or sink	5b

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- **Calculate the differences in pressure at different depths in a liquid (1c, 3c).**
- Calculate relevant values of stored energy and energy transfers; convert between newton-metres and joules (1c, 3c).
- Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system (1a, 1c, 3c).

Suggested practicals

- Investigate the upthrust on objects in different liquids.
- Investigate the stretching of rubber bands.